ENGR 202 Week 7 Project Plan

Lab 6 Characterizing Frequency Dependent Circuits

Member A RC lead

Assigned member: _____ Group: _____

Meeting time: Thursday November 12 and Friday November 13, 2026
6:00-8:00 PM on both days, Pacific Standard Time (Oregon local time).

Responsibility: Q3.1-Q3.7 and Q3.9

Check each item only after its calculation, file, or experimental evidence has been saved.

Before Thursday

☐ A1 Confirm the RC schematic and instrument mapping: R = 1000 ohm, C = 0.22 uF, CH1 = Vin, CH2 = Vout across the capacitor. Share component values with C.

☐ A2 Complete Q3.2-Q3.3: explain low-pass behavior and derive H, magnitude, phase, and cutoff angular frequency with intermediate steps.

☐ A3 Complete Q3.4-Q3.5: substitute values, convert to Hz, check ideal fc = 723.43 Hz, and prepare your own hand-drawn magnitude and phase Bode plots.

Thursday meeting

☐ A4 Verify RC wiring and common ground; save the actual setup photograph for Q3.1.

☐ A5 Lead Q3.6 measurements at all 11 required frequencies, plus additional points near cutoff. Ask B to record values and verify cursor alignment.

☐ A6 Save a complete capture at every point showing both channels, VRMS, Vpp, frequency, and cursor time difference. Record actual frequency and signed phase.

☐ A7 Complete the RC table, gain in dB, measured half-power frequency, and stopband slope with units. Repeat ambiguous or inconsistent points before ending the session.

Friday meeting and final handoff

☐ A8 Complete Q3.7 in Excel: magnitude and phase versus frequency, numeric logarithmic frequency axes, and labeled measured cutoff.

☐ A9 Complete Q3.9 with C: compare theory, hand sketches, LTspice, and measurements; overlay theoretical and measured magnitude in Excel and explain supported discrepancies.

☐ A10 Assist B with RL recording and cursor checks. Deliver the full RC evidence package and approve the RC section after C integrates it into the report.

Required frequencies: 10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000, and 20000 Hz. Add points near cutoff. Use a 2 Vpp sine input.

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Member B RL lead

Assigned member: _____ Group: _____

Meeting time: Thursday November 12 and Friday November 13, 2026
6:00-8:00 PM on both days, Pacific Standard Time (Oregon local time).

Responsibility: Q4.1-Q4.7 and Q4.9

Check each item only after its calculation, file, or experimental evidence has been saved.

Before Thursday

☐ B1 Confirm the RL schematic: R = 690 ohm, L = 0.1 H, output across the inductor. Share values with C; use CH1 = Vin and CH2 = Vout.

☐ B2 Complete Q4.2-Q4.3: explain high-pass behavior and derive H, magnitude, phase, and cutoff angular frequency. Use L despite the worksheet references to C.

☐ B3 Complete Q4.4-Q4.5: convert expressions to Hz, check ideal fc = 1098.17 Hz, and prepare your own hand-drawn magnitude and phase Bode plots.

Thursday meeting

☐ B4 Assist A with RC data recording and same-direction cursor crossings. Flag missing readings, unclear phase signs, and incomplete screenshots.

☐ B5 Prepare the RL wiring for the next session. With power off and the inductor disconnected, measure and record its DC resistance as evidence for a possible nonideal model.

Friday meeting and final handoff

☐ B6 Verify RL wiring and grounding; save the actual setup photograph for Q4.1.

☐ B7 Lead Q4.6 measurements at all 11 required frequencies and extra points near cutoff. Save complete cursor and measurement-window captures for every point.

☐ B8 Complete the RL table, signed phase, gain in dB, measured half-power frequency, and stopband slope. Repeat doubtful measurements.

☐ B9 Complete Q4.7 in Excel with logarithmic numeric frequency axes, magnitude and phase plots, and the measured cutoff.

☐ B10 Complete Q4.9 with C: compare ideal theory, simulation, and experiment. Assess nonideal effects against the measured evidence without presenting a hypothesis as proven.

☐ B11 Deliver the full RL evidence package, place its traces in Appendix C, and approve the integrated RL section.

Required frequencies: 10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000, and 20000 Hz. Add points near cutoff. Use a 2 Vpp sine input.

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Member C Simulation and report integration lead

Assigned member: _____ Group: _____

Meeting time: Thursday November 12 and Friday November 13, 2026
6:00-8:00 PM on both days, Pacific Standard Time (Oregon local time).

Responsibility: Q3.8, Q4.8; coordinate Q1.1 and final review

Check each item only after its calculation, file, or experimental evidence has been saved.

Before Thursday

☐ C1 Confirm group number, member names, leader, actual lab date, and full lab title with A and B for Q1.1. Do not guess missing details.

☐ C2 Build the RC and RL LTspice circuits from the instructor schematics and verify values with A and B. Set source AC amplitude to 1 and sweep .ac dec 100 10 20k.

☐ C3 Complete Q3.8 and Q4.8: save both circuit files, AC settings, magnitude in dB, and phase plots of V(out)/V(in). Record simulated cutoff values.

☐ C4 Prepare shared measurement tables and Excel chart layouts, with units, raw readings, gain, cursor time difference, and signed phase fields.

Thursday meeting

☐ C5 Review the RC captures and table with A: verify all frequencies, RMS/Vpp readings, cursor evidence, phase signs, and gain calculations.

☐ C6 Help A build the RC theoretical/experimental overlay and compare it with simulation. Record unresolved discrepancies and the evidence needed to resolve them.

Friday meeting and final handoff

☐ C7 Perform the same evidence and calculation review for the RL data. Help B compare ideal and measured behavior and assess possible nonideal effects.

☐ C8 Check both Excel workbooks: XY scatter charts, numeric logarithmic frequency axes, dB magnitude, phase in degrees, and clearly labeled cutoff frequencies.

☐ C9 Integrate reviewed content into the original report template. Keep setup photographs with the matching question, RC traces in Appendix B, and RL traces in Appendix C.

☐ C10 Audit all 19 questions: metadata, photographs, derivations, hand sketches, measured tables and cursor captures, Excel plots, simulations, and comparisons. Obtain A and B approval of their sections.

☐ C11 Save the final draft and evidence files in the shared repository. Confirm remaining corrections and the actual Canvas deadline with the designated leader, who handles submission.

Completion standard: Every answer must be supported by the corresponding calculation, measurement, or saved figure. The schedule is a plan; it does not mark experiments as completed.